# Tea and toast: systems and energy

ENGINEERING WITH MR KELLY

National 5 · Engineering contexts & challenges · Class worksheet

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ identify the energy forms going into and out of a product
- ☐ draw universal system and sub-system diagrams
- ☐ describe closed loop control using SET – ACTION – FEEDBACK – COMPARE – ADJUST
- ☐ calculate work done and rearrange to find a mass

## I DO — Key ideas

You will need these

### $E_W = Fd$ and $F = mg$

Work done = force × distance; weight = mass × 9.8 N/kg

- g = 9.8 N/kg

**Remember**

Describe a control system in order: SET – ACTION – FEEDBACK – COMPARE – ADJUST.

## YOU DO — Tasks

Show your working for calculations.

### Toast

**1 State the energy forms for the input and output of a kettle.**

**2 Draw a universal system diagram for a toaster.**

| Input | Process | Output |
|---|---|---|
| | | |

**3 A toaster weighs 15 N. Calculate the work done carrying it up a staircase with a vertical height of 4.2 m.**

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## Tea

**4 Draw a sub-system diagram for an automatic kettle. Include the feedback loop and the system boundary.**

**5 Describe, with reference to your sub-system diagram, the operation of the kettle.**

**6 It takes 58.7 J of work to carry the empty kettle up to the Art department, 7.68 m higher. Calculate the mass of the kettle.** (3 marks)

**7 Describe how the kettle could be made more energy efficient.**

**Check and reflect**

Mark your answers with the answer sheet. Which “I can” statement above still needs work? Look back at the notes for that topic on the site.